

Project Initialization and Planning Phase

Date	27 June 2026
Team ID	
Project Title	Emotion Detection & Learning Support Engine.
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to develop an AI-powered Emotion Detection & Learning Support Engine that identifies students' emotions from text using BiLSTM and BERT models. Based on the detected emotion, the system generates personalized learning guidance using the Gemini API. The solution enhances learning by providing emotional awareness, adaptive recommendations, and an interactive dashboard for students and educators.

Project Overview	
Objective	The primary objective is to build an AI-powered learning support system that detects students' emotions from text and provides personalized learning recommendations to improve engagement and academic performance.
Scope	The project focuses on emotion detection using NLP and deep learning models, AI-generated learning support, emotion analytics, and an interactive Streamlit-based web application.
Problem Statement	
Description	Students often experience stress, frustration, confusion, or low motivation while learning, but existing learning platforms do not consider their emotional state when providing support.

Impact	Addressing this problem will improve student engagement, provide personalized learning assistance, reduce stress, and create a more effective and supportive learning environment.
Proposed Solution	
Approach	The system preprocesses user text, predicts emotions using BiLSTM and BERT models, and generates personalized learning guidance through the Gemini API. Results are displayed on a Streamlit dashboard along with emotion history and analytics.
Key Features	☐ AI-based emotion detection using BiLSTM and BERT. ☐ Personalized learning recommendations using Gemini AI. ☐ Interactive Streamlit dashboard. ☐ Emotion history tracking and analytics. ☐ Confidence score and model comparison.



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	NVIDIA T4 GPU (Kaggle), 2 vCPUs
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	20 GB Available Storage
Software		
Frameworks	Python frameworks	Streamlit, TensorFlow

Libraries	Additional libraries	Transformers, TensorFlow, Pandas, NumPy, NLTK, Scikit-learn, Plotly
Development Environment	IDE	Visual Studio Code, Jupyter Notebook, Kaggle
Data		
Data	Source, size, format	Kaggle Emotion Dataset, CSV format